Charlie Mpetha

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Research Interests

- Galaxy clusters
- Gravitational redshift
- Gravitational wave cosmology

Employment

NASA Postdoctoral Program (NPP) Fellow **Sep 2025 – Present**
NASA Goddard Space Flight Center

Education

Ph.D in Cosmology **Sep 2020 – Apr 2025**
Institute for Astronomy, University of Edinburgh

MSc in Astrophysics **Sep 2019 – Aug 2020**
Astrophysics Research Institute, Liverpool John Moores University

MPhys in Particle Physics and Cosmology **Oct 2015 – Jun 2019**
Department of Physics, Lancaster University

Awards and Funding

2026 Approved Roman Cycle 1 Program, PI's: Mpetha C. T., Choi A.
$150,000 over two years.

2023 Leverhulme Study Abroad Scholarship
£23,040 over one year.

2019 Research Academy Prize

2019 Outstanding Final Year Performance

2018 Cormack/Piazzi Vacation Research Scholarship
£266 per week for six weeks.

Leadership and Service

Referee for *The Astrophysical Journal* **2026**

Galaxy Clusters Deliverables Lead **Jun 2026 – Present**
Roman High Latitude Imaging Survey Project Infrastructure Team

Galaxy Cluster Science Focus Group Co-chair **Feb 2026 – Present**
Roman Science Collaboration

GSFC Roman Science Discussion Organizer **Feb 2026 – Present**

Standard Project Co-Lead **Sep 2025 – Present**
Euclid Collaboration

Large Scale Structure Journal Club Organizer **2022 – 2023**
University of Edinburgh

Publications

First Author

6. Towards Reliably Measuring the Mass-Size Relation of Dark Matter Halos: Impact of projection, miscentering and Eddington bias
Mpetha C. T. et al. 2026, subm. to ApJ, 10.48550/arXiv.2610.36531

5. Cosmology from UNIONS weak lensing profiles of galaxy clusters
Mpetha C. T. et al. 2025, MNRAS, 10.1093/mnras/staf1538

4. The infall region as a complementary probe to cluster abundance
Mpetha C. T. et al. 2024, MNRAS, 10.1093/mnras/stae1637

3. Impact of weak lensing on bright standard siren analyses
Mpetha C. T. et al. 2024, PRD, 10.1103/PhysRevD.110.023502

2. Future prospects of testing extensions to ΛCDM through the weak lensing of gravitational waves
Mpetha C. T. et al. 2023, PRD, 10.1103/PhysRevD.107.103518

1. Gravitational redshifting of galaxies in the SPIDERS cluster catalogue
Mpetha C. T. et al. 2021, MNRAS, 10.1093/mnras/stab544

Significant Co-Author

3. Cluster Properties as a Function of Dynamical State in the DESI Legacy × UNIONS Surveys
Ahad S. L., Reid R., Mpetha C. T. et al. 2026, MNRAS, 10.3847/1538-4357/ae8399

2. Tracing the dynamical states and mass accretion histories of galaxy clusters in IllustrisTNG
Reid R., Ahad S. L., Haggar R., Mpetha C. T. et al. 2026, MNRAS, 10.1093/mnras/stag1093

1. Constraining cosmological parameters using the splashback radius of galaxy clusters
Haggar R., Amoura Y., Mpetha C. T. et al. 2024, ApJ, 10.3847/1538-4357/ad5cee

Other Co-Author

4. Euclid preparation. The shape of halo profiles in ΛCDM and non-standard cosmologies
Pizzuti, L. et al. incl. Mpetha C. T. 2026, subm. to A&A, 10.48550/arXiv.2609.04192

3. Euclid preparation: CV. CosmoPostProcess: A simulation calibrated framework for weak lensing selection bias in richness-selected galaxy clusters
Ingrao R. et al. incl. Mpetha C. T. 2026, A&A, 10.1051/0004-6361/202660776

2. Measuring satellite galaxy subhalo masses in redMaPPer clusters with UNIONS weak lensing data
Akhmetshyn, R. et al. incl. Mpetha C. T. 2026, MNRAS, 10.1093/mnras/stag1318

1. Revisiting the 'Lensing is Low' Problem with UNIONS
Campbell M. C. T. et al. incl. Mpetha C. T. 2026, subm. to MNRAS, 10.48550/arXiv.2606.23651

Teaching and Mentorship

NASA PEER Mentor **Sep 2026 – Present**

Roman JuMP Mentor **November 2025 – Present**
Mentored junior Roman HLIS members, J. Zhang (2025 – 2026) and B. Wells (2026 – Present).

Summer Student Mentor **Jun 2026 – Aug 2026**
Mentored NASA OSTEM Intern Rachel Honor, whose research has led to a publication: *Properties of the Red-Sequence at $z > 1$ for finding Galaxy Clusters*, Honor R., Mpetha C. T. et al., in prep.

Volunteer Tutor **Oct 2024 – Apr 2025**
Volunteer tutoring with the Sudanese Community Group Edinburgh.

Summer Student Mentor **Jul – Aug 2024**
Mentored a high school student on a project visualising simulated galaxy clusters.

Teaching Assistant **Oct 2020 – Present**
- Courses: Cosmology (MSc), Introductory Astrophysics, Modern Physics, Physics 1A+1B (BSc).
- Nominated for *Student Tutor of the Year* in the 2022 Student-Led Teaching Awards.

Maths and Physics Tutor **Dec 2019 – Sep 2023**
- Tutored school students from ages 11–18, and adults, on a range of topics in Maths and Physics.

Invited Presentations

Galaxy Cluster Science with the Roman Space Telescope
University of Pittsburgh AstroLunch (exp.) March 2027

Towards reliably measuring the mass-size relation of dark matter halos
MPE CCG Seminar October 2026

Probing fundamental physics using the gravitational redshift effect in galaxy clusters
LMU Cosmology Lunch Seminar September 2026

Cluster Lensing in the Stage-IV Era
MPIA Seminar September 2026

Dynamical state of galaxy clusters in the Roman COSMOS Deep Field
RSC Cosmology July 2026

Cluster Lensing in the Stage-IV Era
Waterloo Centre for Astrophysics Seminar March 2026

The infall region of galaxy clusters as a complementary probe to cluster abundance
Max Planck Institute for Extraterrestrial Physics January 2025

The infall region of galaxy clusters as a complementary probe to cluster abundance
Yale Center for Astronomy and Astrophysics Cosmology Seminar January 2025

Using the infall region to probe cosmology with UNIONS weak lensing
UNIONS Collaboration weak lensing meeting February & June 2024

Future prospects on testing extensions to ΛCDM through the weak lensing of gravitational waves
LISA Collaboration community meeting January 2023

Contributed Presentations

Probing fundamental physics using the gravitational redshift effect in galaxy clusters
Euclid US Science Review and Team Meeting August 2026

Probing fundamental physics using the gravitational redshift effect in galaxy clusters
RSC Internal Synergies August 2026

Galaxy Cluster Science with the Roman Space Telescope
Goddard SED Director's Seminar July 2026

Delensing Gravitational Waves from LISA Supermassive Black Hole Binaries using the CMB
LISA Consortium Meeting June 2026

Validating the dynamical state of galaxy clusters with lensing
Roman HLIS Cosmology PIT Meeting March 2026

Measuring gravitational lensing of light around galaxy clusters with Roman
MICA Astro-Animation February 2026

Cosmology from lensing profiles of galaxy clusters
Roman HLIS Cosmology PIT Meeting November 2025

The infall region of galaxy clusters as a complementary probe to cluster abundance
KMI Seminar, Nagoya University June 2025

The infall region of galaxy clusters as a complementary probe to cluster abundance
Expanding the Boundaries of Dark Matter Halos, Shanghai Jiao Tong University May 2025

The infall region of galaxy clusters as a complementary probe to cluster abundance
Kavli Institute for Cosmological Physics Survey Science Group Meeting February 2025

The infall region of galaxy clusters as a complementary probe to cluster abundance
MIT Kavli Institute Monday Afternoon Talks February 2025

The infall region of galaxy clusters as a complementary probe to cluster abundance
Harvard-Smithsonian CfA Institute for Theory and Computation Luncheon January 2025

Prospects for using the infall region to probe cosmology
Cosmological and Astrophysical Synergies: Tactics for the Latest Era September 2024

Prospects for using the infall region to probe cosmology
University of Maryland–Visiting speaker August 2024

Using the infall region around galaxy clusters as a cosmological probe
Cosmology from Home June 2024

Using the depletion zone around galaxy clusters as a cosmological test
American Astronomical Society Annual Meeting January 2024

Cosmology from the weak lensing of gravitational waves
Durham-Edinburgh eXtragalactic Workshop XIX January 2023

Cosmology from the weak lensing of gravitational waves
Probing the Universe with Multimessenger Astrophysics September 2022

Statistical cosmology from the weak lensing of gravitational waves
UK Cosmology Meeting May 2022

Posters
CASCA, GWPAW, SUPA Annual Gathering, Tonale Winter School (received poster prize).

Community Engagement

MICA Astro-animation course **Feb 2026 – Jun 2026**
Science Advisor to animation team who produced an animation on the Roman Space Telescope